

# Volume 03 - Book 03.18 Enterprise Governance

Version v20 draft

README.md

## Book 03.18 - Enterprise Governance

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-README

### Purpose

Define the Enterprise Governance blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

### Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Enterprise Governance blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

### Responsibilities

- Establish a canonical blueprint boundary for the Enterprise Governance blueprint package and its subsystem decomposition.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

### Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

### Outputs

- Blueprint contract and subsystem boundaries for the Enterprise Governance blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.

- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- EnterpriseGovernanceDefined
- EnterpriseGovernanceRegistered
- EnterpriseGovernanceStateChanged
- EnterpriseGovernanceReviewRequested
- EnterpriseGovernanceGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.

- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Enterprise Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18 - Enterprise Governance

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-INDEX

## Purpose

Define the Enterprise Governance blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Enterprise Governance blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for the Enterprise Governance blueprint package and its subsystem decomposition.
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- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

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- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for the Enterprise Governance blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
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- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

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- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
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## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
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## Metrics

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## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Enterprise Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.01 - Enterprise Governance Overview

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-001

## Purpose

Define Enterprise Governance Overview within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Enterprise Governance Overview within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation- first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Enterprise Governance Overview within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Enterprise Governance Overview within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- EnterpriseGovernanceOverviewDefined
- EnterpriseGovernanceOverviewRegistered
- EnterpriseGovernanceOverviewStateChanged
- EnterpriseGovernanceOverviewReviewRequested
- EnterpriseGovernanceOverviewGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.



- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Enterprise Governance Overview has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
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| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.02 - Policy Engine

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-002

## Purpose

Define Policy Engine within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Policy Engine within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Policy Engine within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Policy Engine within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- PolicyEngineDefined
- PolicyEngineRegistered
- PolicyEngineStateChanged
- PolicyEngineReviewRequested
- PolicyEngineGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Policy Engine has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.03 - Approval Engine

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-003

## Purpose

Define Approval Engine within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Approval Engine within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Approval Engine within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Approval Engine within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- ApprovalEngineDefined
- ApprovalEngineRegistered
- ApprovalEngineStateChanged
- ApprovalEngineReviewRequested
- ApprovalEngineGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Approval Engine has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

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- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.04 - Exception Management

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-004

## Purpose

Define Exception Management within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Exception Management within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Exception Management within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Exception Management within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces



- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- ExceptionManagementDefined
- ExceptionManagementRegistered
- ExceptionManagementStateChanged
- ExceptionManagementReviewRequested
- ExceptionManagementGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
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- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

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- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

### Acceptance Criteria

- Exception Management has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

### Traceability

- MAL-V00-B009
- MAL-V00-B010
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- MAL-V03-B03-05-003

### Version History

| Version   | Date       | Status | Notes                                                                                |
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| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.05 - Audit System

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-005

## Purpose

Define Audit System within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Audit System within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Audit System within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Audit System within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- AuditSystemDefined
- AuditSystemRegistered
- AuditSystemStateChanged
- AuditSystemReviewRequested
- AuditSystemGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Audit System has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.06 - Risk Register

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-006

## Purpose

Define Risk Register within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Risk Register within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Risk Register within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Risk Register within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- RiskRegisterDefined
- RiskRegisterRegistered
- RiskRegisterStateChanged
- RiskRegisterReviewRequested
- RiskRegisterGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Risk Register has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |



# Book 03.18.07 - Compliance Evidence

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-007

## Purpose

Define Compliance Evidence within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Compliance Evidence within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Compliance Evidence within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Compliance Evidence within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- ComplianceEvidenceDefined
- ComplianceEvidenceRegistered
- ComplianceEvidenceStateChanged
- ComplianceEvidenceReviewRequested
- ComplianceEvidenceGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Compliance Evidence has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.08 - Legal Controls

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-008

## Purpose

Define Legal Controls within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Legal Controls within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Legal Controls within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Legal Controls within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- LegalControlsDefined
- LegalControlsRegistered
- LegalControlsStateChanged
- LegalControlsReviewRequested
- LegalControlsGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Legal Controls has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.09 - Security Governance

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-009

## Purpose

Define Security Governance within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Security Governance within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Security Governance within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Security Governance within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- SecurityGovernanceDefined
- SecurityGovernanceRegistered
- SecurityGovernanceStateChanged
- SecurityGovernanceReviewRequested
- SecurityGovernanceGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.



- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Security Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.10 - Quality Governance

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-010

## Purpose

Define Quality Governance within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Quality Governance within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Quality Governance within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Quality Governance within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- QualityGovernanceDefined
- QualityGovernanceRegistered
- QualityGovernanceStateChanged
- QualityGovernanceReviewRequested
- QualityGovernanceGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

### Acceptance Criteria

- Quality Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

### Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

### Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.11 - Finance Governance

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-011

## Purpose

Define Finance Governance within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Finance Governance within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Finance Governance within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Finance Governance within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- FinanceGovernanceDefined
- FinanceGovernanceRegistered
- FinanceGovernanceStateChanged
- FinanceGovernanceReviewRequested
- FinanceGovernanceGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Finance Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.12 - Governance Reporting

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-012

## Purpose

Define Governance Reporting within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Reporting within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Reporting within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Reporting within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces



- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceReportingDefined
- GovernanceReportingRegistered
- GovernanceReportingStateChanged
- GovernanceReportingReviewRequested
- GovernanceReportingGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

### Acceptance Criteria

- Governance Reporting has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

### Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

### Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.13 - Governance Checkpoints

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-013

## Purpose

Define Governance Checkpoints within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Checkpoints within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Checkpoints within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Checkpoints within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceCheckpointsDefined
- GovernanceCheckpointsRegistered
- GovernanceCheckpointsStateChanged
- GovernanceCheckpointsReviewRequested
- GovernanceCheckpointsGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Checkpoints has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.14 - Governance Analytics

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-014

## Purpose

Define Governance Analytics within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Analytics within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Analytics within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Analytics within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceAnalyticsDefined
- GovernanceAnalyticsRegistered
- GovernanceAnalyticsStateChanged
- GovernanceAnalyticsReviewRequested
- GovernanceAnalyticsGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Analytics has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |



# Book 03.18.15 - Governance Operations

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-015

## Purpose

Define Governance Operations within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Operations within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Operations within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Operations within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceOperationsDefined
- GovernanceOperationsRegistered
- GovernanceOperationsStateChanged
- GovernanceOperationsReviewRequested
- GovernanceOperationsGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Operations has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.16 - Governance Events

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-016

## Purpose

Define Governance Events within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Events within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Events within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Events within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceEventsDefined
- GovernanceEventsRegistered
- GovernanceEventsStateChanged
- GovernanceEventsReviewRequested
- GovernanceEventsGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Events has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.17 - Governance Storage

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-017

## Purpose

Define Governance Storage within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Storage within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Storage within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Storage within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceStorageDefined
- GovernanceStorageRegistered
- GovernanceStorageStateChanged
- GovernanceStorageReviewRequested
- GovernanceStorageGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.



- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Storage has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.18 - Governance Security

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-018

## Purpose

Define Governance Security within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Security within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Security within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Security within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceSecurityDefined
- GovernanceSecurityRegistered
- GovernanceSecurityStateChanged
- GovernanceSecurityReviewRequested
- GovernanceSecurityGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

### Acceptance Criteria

- Governance Security has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

### Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

### Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.19 - Governance Testing

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-019

## Purpose

Define Governance Testing within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Testing within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Testing within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Testing within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceTestingDefined
- GovernanceTestingRegistered
- GovernanceTestingStateChanged
- GovernanceTestingReviewRequested
- GovernanceTestingGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Testing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |

# Book 03.18.20 - Governance Roadmap

Version: v20 draft

Status: Draft

Document ID: MAL-V03-B03-18-020

## Purpose

Define Governance Roadmap within the Enterprise Governance blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

## Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Governance Roadmap within the Enterprise Governance blueprint. It does not implement product code or bypass the documentation-first governance model.

## Responsibilities

- Establish a canonical blueprint boundary for Governance Roadmap within the Enterprise Governance blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

## Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Enterprise Governance.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

## Outputs

- Blueprint contract and subsystem boundaries for Governance Roadmap within the Enterprise Governance blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

## Interfaces



- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

## Events

- GovernanceRoadmapDefined
- GovernanceRoadmapRegistered
- GovernanceRoadmapStateChanged
- GovernanceRoadmapReviewRequested
- GovernanceRoadmapGovernanceBlocked

## Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

## Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

## Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

## Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

## Acceptance Criteria

- Governance Roadmap has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

## Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B018
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

## Version History

| Version   | Date       | Status | Notes                                                                                |
|-----------|------------|--------|--------------------------------------------------------------------------------------|
| v20 draft | 2026-06-29 | Draft  | Initial Enterprise Governance blueprint content for Mariam Architecture Library v20. |